Project Report: Travel and Tourism Management System

1. Project Overview

The **Travel and Tourism Management System** is a comprehensive Java-based application designed to automate the functioning of a travel agency. It provides a user-friendly interface for customers to manage their profiles, explore travel packages, book hotels, and handle payments seamlessly.

2. Technology Stack

Component	**Technology**
Programming Language	Java (JDK 25)
GUI Framework	Java Swing & AWT
Database	MySQL Server
IDE	IntelliJ IDEA 2025.3.1.1
Database Connector	MySQL Connector/J 9.6.0

3. System Architecture

The application follows a modular architecture where the front-end (Java Swing) communicates with the back-end (MySQL) via JDBC (Java Database Connectivity).

Database Schema

The system utilizes several interconnected tables to maintain data integrity:

1. **account**: Stores user login credentials and security details.
2. **customer**: Stores detailed personal profiles linked to accounts.
3. **hotel**: Contains base rates and metadata for various accommodations.
4. **bookpackage**: Records details of travel packages booked by users.
5. **bookhotel**: Records specific hotel bookings including AC and food preferences.

4. Key Features

- **Authentication Module**: Secure Login and Signup with mandatory field validation and duplicate username checks.
- **Personal Profile Management**: Modules for adding, viewing, updating, and deleting personal details.
- **Dynamic Booking**:
    - **Package Booking**: Automated price calculation for Gold, Silver, and Bronze packages.
    - **Hotel Booking**: Real-time cost estimation based on hotel choice, number of days, AC selection, and food preferences.
- **Visual Galleries**: Interactive slideshows for checking hotels and exploring travel destinations.
- **Utilities**: Built-in access to system tools like Calculator and Notepad.
- **Security**: "Forget Password" functionality driven by security questions.

5. Implementation Details

The project is implemented using advanced Java concepts including:

- **Multithreading**: Used for the smooth execution of visual slideshows without blocking the main UI thread.
- **JDBC**: Used for executing CRUD (Create, Read, Update, Delete) operations.
- **Event Handling**: Robust `ActionListener` and `MouseListener` implementations for an interactive User Interface.

6. Project Impact

This system significantly reduces manual paperwork and improves the efficiency of travel bookings. For academic purposes at **IIT (ISM) Dhanbad**, it demonstrates a strong integration of Data Analytics concepts with practical Software Engineering and Database Management practices.

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